
Physics-Gated Correction Discovery: Why Constraining the Search Space Beats Scaling It in Noisy Symbolic Regression

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Abstract

Symbolic regression (SR) methods such as PySR, AI Feynman, and PhySO discover equations *from scratch*, an approach that is sensitive to noisy physical data: under matched residuals, PySR’s structural accuracy is non-monotonic in noise and does not improve when its search budget is doubled. We argue that scientific discovery is rarely *tabula rasa*; it is *correction-first*. Given a known classical law and anomalous data, the real task is to find the minimal physically-valid correction Δ . We present **Anomaly-Driven Correction Discovery (ADCD)**, which restricts search to a correction subspace and cascades three physical filters (AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency [ARC]) *before* a JAX optimizer fits any parameter, with Bayesian Information Criterion (BIC) reranking. On a 9-scenario benchmark (overall mean $80.4\% \pm 7.4\%$ across 16 seeds and all noise levels; 88.9% at the disclosed reference seed), ADCD reaches 88.9% at 5% noise versus 11.1% for PySR, a *seed-independent* gap: ADCD’s *worst* of 16 seeds still exceeds PySR with doubled budget. Removing all gates drops recovery by 44.5 pp. ADCD is orthogonal to, not competitive with, existing SR: it restricts the problem they solve to a physically-motivated subspace where unconstrained search is least reliable under noise.

1 Introduction

When a physical law breaks down under extreme conditions, the scientific question is not “what is the correct law?” Instead, it is “what is the *minimal correction* to the known law?” The dominant paradigm in symbolic regression (SR) answers a different question: it discovers $f(X) \approx y$ from a blank slate, an approach that, as we show, degrades under realistic noise. Pioneered by Eureqa [Schmidt & Lipson \[2009\]](#), AI Feynman [Udrescu & Tegmark \[2020\]](#), and PySR [Cranmer \[2020\]](#), and extended by deep RL approaches such as DSR [Petersen et al. \[2021\]](#) and unit-constrained PhySO [Tenachi et al. \[2023\]](#), these systems recover compact laws in *low-noise* regimes. Under realistic observational noise, however, their structural accuracy degrades and is often *non-monotonic* in noise: a larger candidate pool does not reliably translate into the correct functional *class* (Section 4).

This difficulty stems in part from a limitation in *problem formulation*. Science rarely discovers from a blank slate; it *corrects*. From the relativistic kinetic-energy expansion $\frac{1}{2}mv^2(1 + \frac{3}{4}v^2/c^2 + \dots)$, to Debye-screened Coulomb potentials $V(r) \propto r^{-1}e^{-r/\lambda_D}$, to turbulent drag $F = bv + \alpha v^2$, progress takes a structurally-correct classical law and appends a *minimal correction* Δ that reconciles it with anomalous data. This restricts the hypothesis space from the full function space $\mathcal{F}$ to a correction subspace $\mathcal{D} \subset \mathcal{F}$, an exponential reduction informed by the known physics.

The proposed **Anomaly-Driven Correction Discovery (ADCD)** framework introduces three main contributions: **(C1) Correction-first paradigm**: Search for Δ in a restricted correction subspace

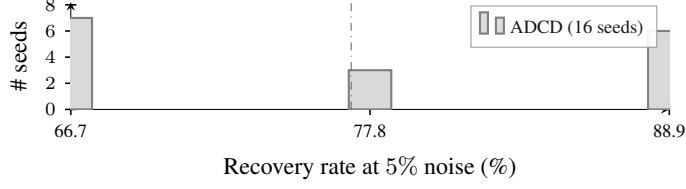
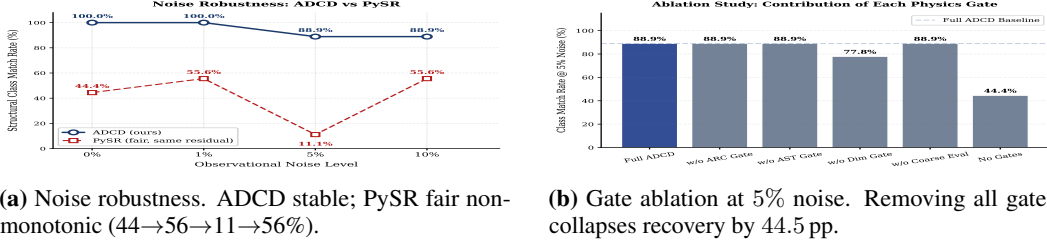


Figure 1: Seed-independence of the ADCD-PySR gap at 5% noise. Every one of ADCD’s 16 seeds (bars) recovers at least 66.7%, above PySR’s best result even under doubled budget (55.6%, dotted).



(a) Noise robustness. ADCD stable; PySR fair non-monotonic (44→56→11→56%).

(b) Gate ablation at 5% noise. Removing all gates collapses recovery by 44.5 pp.

Figure 2: Left: structural match rate vs. noise. Right: per-gate ablation.

$\mathcal{D} \subset \mathcal{F}$, not the full function space f in $\mathcal{F}$, given a known classical law $y_{\text{classical}} = g(X; \phi)$ and anomalous observations. **(C2) Physics-gated cascade before optimization:** Three physical consistency filters (AST complexity, dimensional homogeneity, and asymptotic decoupling limits $\Delta \rightarrow 0$ at the classical boundary) screen candidates *before* any parameter is numerically fit. **(C3) Noise-robust stability:** ADCD’s structural accuracy is monotonic in noise because it avoids the universal-approximator regime that overfits noise into nested mathematical structures.

We deliberately scope this paper to *method and synthetic benchmark*; real-data validation (galactic, cosmological) is reported in a companion manuscript.

2 Related work

Tabula rasa symbolic regression. Modern SR discovers an expression $f(X) \approx y$ from a blank slate. Eureka Schmidt & Lipson [2009] pioneered genetic search over expression trees; AI Feynman Udrescu & Tegmark [2020] added concepts like symmetry and unit filtering; PySR Cranmer [2020] scaled genetic SR; DSR Petersen et al. [2021] applied reinforcement learning. All share a common limitation: under realistic noise their structural accuracy degrades and is often *non-monotonic* in noise, because a larger search space rewards overfit structure rather than the correct functional class. ADCD departs by searching only the correction subspace $\mathcal{D} \subset \mathcal{F}$ to a *known* law, an exponential restriction of the hypothesis space.

Unit-constrained & LLM-guided SR. PhySO Tenachi et al. [2023] enforces dimensional homogeneity during RL training, while LaSR Grayeli et al. [2024] uses language models to evolve a concept library. ADCD is complementary: its physics-gated cascade (dimensional, asymptotic, complexity) filters proposed candidates *before* any numerical parameter optimization, regardless of whether they are generated by grammar or LLM samplers.

Physics in models vs. search space. Instead of baking physics into the loss (PINNs Raissi et al. [2019]), coordinates (HNNs Greydanus et al. [2019]), or differential equations (SINDy Champion et al. [2019]), ADCD restricts the symbolic *search space* directly. It provides interpretable, pre-fit filters that any downstream optimizer operates within.

3 The ADCD framework

Residual extraction. Given a known classical law $y_{\text{classical}} = g(X; \phi)$ and noisy anomalous data y_{obs} , ADCD isolates the correction by forming a residual. In *multiplicative* mode $y_{\text{true}} =$

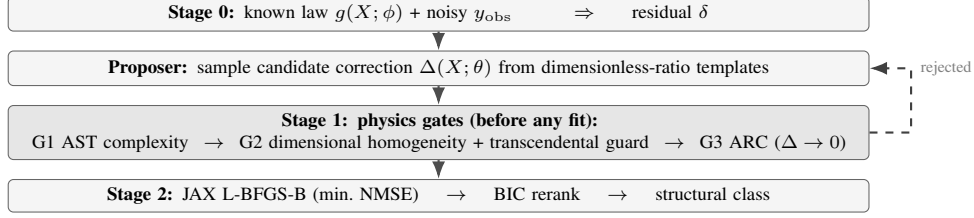


Figure 3: ADCD pipeline. Physics gates (G1–G3) screen candidates *before* any parameter is fit; rejected candidates are re-sampled.

$y_{\text{classical}}(1 + \Delta)$ gives $\delta = y_{\text{obs}}/y_{\text{classical}} - 1$; in *additive* mode $y_{\text{true}} = y_{\text{classical}} + \Delta$ gives $\delta = y_{\text{obs}} - y_{\text{classical}}$. A mode selector picks whichever best isolates the anomalous deviation.

Proposer. Candidate correction templates $\Delta(X; \theta)$ are drawn from a stochastic sampler over dimensionless ratios, biased by residual features (monotonicity, curvature, oscillation, exponential decay R^2 , symmetry) that upweight plausible families but never gate candidates; only the physics cascade does.

Stage 1: cascaded physics gates. Every candidate passes three physical filters *before* parameter fitting: **G1. AST complexity:** Reject expressions with AST depth > 7 or > 20 tokens (Occam prior). **G2. Dimensional homogeneity:** Verify Δ is dimensionless; physical variables enter only inside dimensionless ratios, and transcendental arguments (e.g., inside $\exp$, $\sin$) must be dimensionless. **G3. Asymptotic-regime consistency (ARC):** Enforce the decoupling limit: $\lim_{x \rightarrow x_0} \Delta(X; \theta) = 0$ (e.g., $v \rightarrow 0$ or $r \rightarrow \infty$) evaluated via SymPy. Survivors are coarsely ranked by preliminary NMSE on δ , giving an early data-driven signal before the full fit.

Stage 2: JAX-traced optimization + BIC reranking. Surviving expressions are transpiled to JAX and fit by L-BFGS-B (multi-restart, log-uniform init) minimizing NMSE of the residual: $\text{NMSE} = \frac{\sum_i (\Delta(X_i; \theta) - \delta_i)^2}{\sum_i (\delta_i - \delta)^2}$. Under noise, flexible wrong-class expressions can achieve lower NMSE than the truth, so we rerank by the Bayesian Information Criterion [Schwarz \[1978\]](#), [Rissanen \[1978\]](#): $\text{BIC} = N \ln(\text{NMSE}) + k \ln N$, whose $k \ln N$ penalty selects parsimonious correct-class forms. The winner is classified into one of six families by AST traversal, with degenerate-exponent detection.

Relation to EFT. The cascade mirrors the correction-first philosophy of Effective Field Theory [Weinberg \[1979\]](#): dimensional analysis of operators (G2), decoupling of heavy modes in the low-energy limit (G3/ARC), and power counting (BIC). ADCD has no renormalization-group flow or controlled E/Λ expansion; instead, it *automates* that philosophy for data-driven problems.

4 Experiments

Benchmark. We evaluate on 9 physical-anomaly scenarios in three tiers: *textbook* (relativistic KE, Yukawa gravity, anharmonic spring), *cross-domain* (net radiation T^{-4} , nonlinear drag, screened Coulomb), and *synthetic novel* formulas. Each is run at 0/1/5/10% observational noise. Success = correct structural *class* recovery (e.g. discovering an exponential correction where the truth is exponential). We report a 16-seed distribution; the mean is the primary claim and seed=42 (the highest seed) is disclosed explicitly, never as the headline.

Baselines. We compare against PySR [Cranmer \[2020\]](#) on the *same residual targets* $\delta = y_{\text{obs}} - y_{\text{classical}}$. The primary baseline is the *fair* profile ($n_{\text{iter}}=100$, $\text{maxsize}=30$, 60 s/scenario). To rule out a budget artefact we add a *generous* profile that *doubles* every budget axis ($n_{\text{iter}}=200$, $\text{maxsize}=40$, 120 s; $\sim 3 \times$ wall-clock).

Main result: the gap is structural, not budgetary. Table 1 reports structural match rates. ADCD reaches 88.9% (8/9) at 5% noise versus 11.1% (1/9) for PySR *fair*, representing a 77.8 pp gap under matched residuals. The comparison is *seed-independent*: at 5% noise ADCD’s worst of 16 seeds (66.7%, 6/9; cf. the per-seed distribution in our bundled artefacts) still exceeds PySR *fair*. Doubling PySR’s budget does *not* close the gap; generous total recovery stays flat at 15/36 (identical to *fair*)

and at the 5% headline point it reaches only 5/9 (55.6%). The extra budget explores deeper trees (mean complexity 9.8 vs. 8.7 nodes) spent on more deeply-overfit rather than more physically-correct structure. The binding constraint is *structural selection*, not search budget; physics gates resolve it directly.

Table 1: Structural class match rate on the 9-scenario residual benchmark. *generous* doubles PySR’s budget on every axis yet does not close the gap.

Method	0%	1%	5%	10%
ADCD (ours, seed=42)	9/9 (100%)	9/9 (100%)	8/9 (88.9%)	8/9 (88.9%)
ADCD 16-seed mean	86.8%	81.3%	$77.1\% \pm 10.0\%$	76.4%
PySR fair	4/9 (44.4%)	5/9 (55.6%)	1/9 (11.1%)	5/9 (55.6%)
PySR generous (2×)	4/9 (44.4%)	4/9 (44.4%)	5/9 (55.6%)	2/9 (22.2%)

The seed-independence of this gap is structural, not artefactual. Figure 1 bins ADCD’s per-seed recovery at the 5% headline noise level: the entire 16-seed distribution sits between 66.7% and 88.9%, *entirely above* PySR’s best result even under doubled budget (55.6%, *generous*). ADCD’s worst seed exceeds PySR’s best by 11.1 pp; the gap cannot be attributed to a favourable seed choice.

Why PySR is non-monotonic, and why ADCD is not. PySR fair’s accuracy is non-monotonic in noise (4, 5, 1, 5 out of 9) because at 0% noise it overfits to complex trigonometric structures (mean 13.8 nodes). ADCD avoids this because its physical gates are noise-independent: dimensional and ARC filters reject overfit structures before optimization, allowing robust selection across all SNRs (Figure 2a).

Ablation: the gates act collectively. Disabling gates one at a time at 5% noise (Figure 2b) shows that removing ARC, AST, or the coarse-eval pre-filter leaves accuracy unchanged (8/9), while disabling the dimensional gate alone costs 11.1 pp (8/9 $\rightarrow$ 7/9). Disabling all gates collapses performance (8/9 $\rightarrow$ 4/9, −44.5 pp), showing that the gates act collectively to enforce physical consistency.

5 Discussion and limitations

ADCD is a restriction, not a competitor. ADCD does not compete with PhySO or PySR; it restricts the problem they solve to a physically-motivated subspace where unconstrained search is least reliable under noise. This makes ADCD directly *combinable* with them: a PhySO/PySR/LaSR Grayeli et al. [2024] proposer could feed the same gate cascade.

Honest limitations and proposer template bias. ADCD assumes a structurally-correct classical baseline (incremental correction, not wholesale replacement). Under high noise ($\geq 5\%$), screened-Coulomb decay is fitted by a rational proxy, reflecting decay-vs-saturation ambiguity under limited dynamic range. Crucially, the current proposer layer (whether grammar- or LLM-driven) is fundamentally bottlenecked by *prior template bias*: it samples candidate corrections from a predefined dictionary of known functional families. This restricts the search space to mathematical structures already documented in human physics literature, leaving the framework unable to discover truly unexpected or non-templated functional corrections.

The breakthrough path: complete functional bases. To escape this template bottleneck, our ongoing work focuses on replacing heuristic templates with a systematic, non-templated expansion. By applying the Buckingham-II theorem to the system variables, we construct independent dimensionless groups ($\Pi_1, \Pi_2, \dots$), and expand the correction Δ directly in a *complete functional basis* (e.g., Chebyshev polynomials, Legendre polynomials, or Bessel series) defined over these groups. This shifts the paradigm from "rediscovering known families" to a systematic mathematical search over complete function spaces, ensuring that unexpected physics can be captured without prior bias.

Reproducibility and fail-safe behavior. ADCD ships as an open-source Python package with a JAX backend, public CLI, and bundled test seeds. Unlike GPU-intensive frameworks, it runs in seconds on consumer CPUs, making replication accessible without HPC resources. On a 6-case

misspecification benchmark, it fails to converge rather than outputting confident spurious equations, establishing a crucial fail-safe property.

AI disclosure. Large language models assisted in code development and manuscript drafting under human guidance. The human author conceived the hypotheses, designed the experiments, verified all findings, and remains fully responsible for the work.

Closing. Noisy symbolic regression is bottlenecked by *structural selection*, not search budget. The correction-first formulation, which searches for a physically-valid Δ to a known law screened by dimensional and asymptotic gates before any fit, resolves this directly: ADCD delivers seed-independent gains over PySR, stays robust where unconstrained search is non-monotonic, and is orthogonal to existing SR systems. Future work extends the cascade to multivariable laws and integrates it as a selection layer over external proposers (PhySO, LaSR).

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